



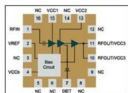
Each channel contains two output stages, one for pulsed wave (PW) operation, together with an integrated transmit/receive switch to guarantee effective decoupling during the transmission phase. The STHV800's fast and symmetrical output voltage transitions allow sharp images to be obtained without the need for complex and costly image-processing algorithms.

Key features of the STHV800 include an input voltage range of 1.5V to 3.6V and an output voltage up to $\pm 90V$, the ability to control pulses with durations as low as 10ns for maximum image sharpness, less than 10ps jitter to ensure accurate frequency response in ultrasound Doppler analysis, low second-harmonic distortion (less than -40dB) for higher image quality, and a design that does not need external reference capacitors. Housed in a compact LGA8x8 package, the STHV800 is now available in volume at a price of \$16 in quantities of 1000 units.

STMicroelectronics, Noida
Phone: 0120-2352999
Website: www.st.com

RF high-power amplifier

Microchip has announced its latest 2.4GHz 256-QAM RF high-power amplifier, the SST12CP21, which offers ultra-low EVM and current consumption for 256-QAM and IEEE 802.11n systems. The SST12CP21 delivers high linear output power of up to 23dBm at 1.75% dynamic EVM, with MC59 HT40 MHz bandwidth modulation at 5V and 320mA current consumption. Additionally, the SST12CP21 delivers 25dBm linear power at 3% EVM with only 350mA current consumption for 802.11g/n applications. This performance significantly extends the range of 802.11b/g/n WLAN and MIMO systems, while consuming extremely low current at the maximum 256-QAM data rate. The



SST12CP21 is also spectrally compliant up to 28dBm for 802.11b/g

communication. Board space is reduced by the small 3x3x0.55mm, 16-pin QFN package that matches a popular pin-out.

The SST12CP21 power amplifier has a low operating current of 320mA at 23dBm and 350mA at 25dBm, which enables multichannel and higher data rate WLAN systems. This amplifier also features 50-ohm on-chip input match and simple output match, which is easy to use and reduces board size. Additionally, the integrated linear power detector provides accurate output power control over temperature and 2-to-1 output mismatch.

The SST12CP21 is available for sampling and volume production, in the popular 3x3x0.55mm, 16-pin QFN package. Pricing is \$0.754 each, in 10,000-unit quantities.

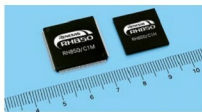
Microchip, India
Phone: 080-30904444; 011-41608631
E-mail: asia.inquiry@microchip.com

Microcontrollers for motor control

Renesas Electronics Corporation has announced the new RH850/C1x series of 32-bit microcontrollers (MCUs), designed for motor control in hybrid electric vehicles (HEVs) and electric vehicles (EVs). Based on Renesas Electronics' 40nm process, the RH850/C1x series features the RH850/CIH and RH850/CIM MCUs, which enable embedded designers to enhance efficiency, reduce system costs and achieve higher safety levels for HEV/EV motor control systems.

The RH850/C1x can be used with the RAA27000KFT RH850 family power supply management IC, which is currently available in sample quantities. The power management IC integrates into one device all the power supply systems required for MCU operation, two external sensor power supply tracks and a full complement of monitoring and diagnostic functions, significantly reducing the user burden associated with power supply system design.

By integrating hardware peripherals dedicated for motor control into the MCU, designers can reduce overall system costs and meet performance requirements. HEV/EV systems are



now being designed to meet ISO 26262 functional safety standards. This system requirement is driving semiconductor products to incorporate a number of embedded features that allow system designers to meet their safety goals.

Renesas Electronics India Pvt Ltd, India
Phone: +91-8067208700; 011-66207300
Website: <http://in.renesas.com>

MISCELLANEOUS

Liquid level indicator & controller

Ascem has launched multifunction digital liquid level indicator and controller model LMC-01 based on a microcontroller. It has been indigenously designed, tooled and manufactured by Ascem with the help of their R&D department.

The equipment has a timer alarm and sensor-fail indicator. It eliminates the need of levelling the tank and makes the control of motor and pump convenient.

Ascem System, Vadodra
Phone: 0265-3022222
E-mail: sales@ascemindia.com

Solar-powered products

SunSwitch India has introduced modular solar home lighting systems and solar retrofits for home inverters to cater to the needs of both urban and rural markets in India.

AnantUrja (12V) and Surya Dhani (3.25V) are the two solar home lighting systems based on Li-ion battery. A customer can invest in pieces of the modular systems and add further modules, when needed. Pakshak is a solution for retrofitting home inverters with solar charging facility.

SunSwitch India (P) Ltd, Sahibabad (UP)
Phone: 1800-11-8090
E-mail: info@sunswitch.in